INCH-POUND

A-A-59684
19 October 2004

COMMERCIAL ITEM DESCRIPTION

OPTICAL SPECTRUM ANALYZER

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Purchase Description (PD) describes an Optical Spectrum Analyzer in which only those manufacturers meeting or surpassing the following requirements are qualified per this PD. This instrument measures signals with a 10 pm resolution and a 20 pm wavelength accuracy in the 1.55 μmeter band.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and bid sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification: Equipment covered by this PD may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 4 in accordance with MIL-PRF-28800.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.3 Mean Time Between Failures (MTBF). The equipment Mean Time Between Failures (MTBF) shall be a minimum of 8,000 hours.

2.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

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2.5 Performance Characteristics.

2.5.1 Salient Features. This instrument shall have the following salient features.

2.5.2 Measurement Range. This instrument shall have the following measurement range: 600 to 1700 nm.

2.5.3 Measurement Accuracy. This instrument shall have the following measurement accuracy, with a built in calibration light source and the readings shall be taken at 23 °C +/- 5°C at 1 minimum repetition rate: ≤ +/- 20 pm. This specification applies to the 1530 nm to 1570 nm band.

2.5.4 Waveform Resolution. This instrument shall have the following waveform resolution settings: 10 pm, 20 pm, 50 pm, 100 pm, and 500 pm.

2.5.5 Waveform Resolution Accuracy. This instrument shall have the following waveform resolution accuracy: ≤ +/- 3% (with a resolution of 50 pm over the 1530 nm to 1610 nm range). Alternately, this instrument shall have the following waveform resolution accuracy: ≤ +/- 2% (with a resolution of 100 pm or greater over the 1530 nm to 1610 nm range).

2.5.6 Measurement Level Range. This instrument shall have the following measurement level range under the following two conditions A) measurements taken at 10 °C to 30 °C and B) with at least 100 pm resolution: -87 to +23 dBm (1250 nm to 1610 nm) and -77 to +23 dBm (950 nm to 1250 nm).

2.5.7 Measurement Level Accuracy. This instrument shall have the following measurement level accuracy under the following two conditions A) at 23 °C +/- 5 °C and B) with at least 100 pm resolution: ≤ +/- 0.4 dB (at a wavelength of 1550 nm).

2.5.8 Measurement Level Linearity. This instrument shall have the following measurement level linearity under the following condition A) measurements taken at 23 °C +/- 5 °C: ≤).05 dB (-50 to 10 dBm, at a wavelength of 1550 nm).

2.5.9 Measurement Level Scale. This instrument shall have the following measurement level scale: a linear scale and a logarithmic scale of: 0.1, 0.2, 0.5, 1, 5 and 10 dB/division.

2.5.10 Measurement Level Repeatability. This instrument shall have the following measurement level repeatability taken under the following three conditions: A) measurements taken at 23 °C +/- 5 °C B) with at least 100 pm resolution and C) measurements taken at a minimum repetition rate of 1: ≤ +/- 0.02 dB from 1530 to 1610 nm wavelengths.

2.5.11 Measurement Level Flatness. This instrument shall have the following measurement level flatness taken under the following condition: A) measurements shall be taken at 23 °C +/- 5 °C. This flatness shall be: ≤ +/- 0.2 dB from 1530 to 1610 nm wavelengths.

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2.5.12 Measurement Level Polarization Dependency. This instrument shall have the following measurement level polarization dependency taken under the following two conditions: A) measurements shall be taken at 23 °C +/- 5 °C and B) measurements shall be taken with at least 100 pm resolution. This flatness shall be: ≤ +/- 0.05 dB from 1250 to 1610 nm wavelengths under conditions A) and B).

2.5.13 Measurement Level Dynamic Ranges. This instrument shall have the following measurement level dynamic ranges taken under the following two conditions: A) measurements shall be taken at 23 °C +/- 5 °C and B) measurements shall be taken with at least 10 pm resolution and at 1523 nm. This dynamic range shall be: ≥ 50 dB (+/- 0.1 nm from the peak wavelength under conditions A) and B). This dynamic range shall be: ≥ 60 dB (+/- 0.2 nm from the peak wavelength under conditions A) and B). This dynamic range shall also be: ≥ 67 dB (+/- 0.4 nm from the peak wavelength under conditions A) and B).

2.6 Sweep Characteristics – Span. This instrument shall have the following sweep span: 0.2 nm from full span to zero span.

2.6.1 Sweep Characteristics – Number of Samples. This instrument shall have the following sweep span number of samples: 101, 201, 501, 1001, 2001, 5001 and 10001.

2.6.2 Sweep Characteristics – Measurement Time. This instrument shall have the following sweep measurement times: ≤ 500 ms (with a span of 10 nm, in the normal mode, and 1550 nm, with an average of 1 time span and with 501 samples).

2.7 Pulsed Light Measurements – Peak Holding Mode. This instrument shall have the following pulsed light measurements - Peak Holding Mode: The waiting time shall be set at every measurement point with a gate time of 1ms to 1 second.

2.7.1 Pulsed Light Measurements – External Synchronization. This instrument shall have the following pulsed light measurements - External Synchronization: the ability to have a sync pulse at the external input control the taking of a reading/measurement. It shall be a TTL logic high level of 3.5 volts with a TTL logic low level of 1.5 volts constituting a logic level low or inhibit function.

2.7.1.1 Pulsed Light Measurements – External Synchronization (Sync Hi Mode). This instrument shall have the following pulsed light measurements - External Synchronization: the ability to have a TTL logic high level of 3.5 volts with this TTL logic high level function causing sample timing to occur from the rising to falling pulse edge, and be set by the SYNC signal and ranges from 0 to 1,000 μsecs.

2.7.1.2 Pulsed Light Measurements – External Synchronization (Sync Low Mode). This instrument shall have the following pulsed light measurements - External Synchronization: the ability to have a TTL logic low level of 1.5 volts level function be when sample timing occurs from the falling to rising pulse edge, and be set by the SYNC signal and with the pulse width being 1,000 nanoseconds or greater.

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2.8 Internal Memory. This instrument shall have an Internal Memory in the form of RAM with a capacity, as a minimum, of 501 samples and at least 15 screen shots on the display.

2.9 Optical Input. This instrument shall have an optical input of an adapting fiber 9.5/125 µmeters of SM fiber with a grade-A connector.

2.9.1 Optical Input Reflective Attenuation. This instrument shall have an optical input Reflective Attenuation of ≥ 35 dB.

2.10 Optical Data Input/Output. This instrument shall have an optical data input/output of a GPIB IEEE-488 standard, as well as a D-SUB 25 pin ESC/P, ESC/P-R, and PCL standards.

2.11 Light Source for Wavelength Calibration. This instrument shall have an optical light source for wavelength calibration of any technology as long as there is an output level of at least ≥ 45 dBm/nm at a wavelength of 1550 nm.

2.12 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 The contracting officer has the option of accepting or rejecting the product.

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5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 IEEE standards. IEEE standards may be obtained by contacting IEEE Operations Center, 445 Hoes Lane, PO Box 1331, Piscataway, NY, 08854-1331. Phone 800-678-4333 (Toll-Free, US & Canada), 732-981-0060, Fax: 732 981 9667 www.ieee.org.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

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